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**Лабораторная работа № 2 по курсу
«Операционные системы и архитектура компьютера»**

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Условие

Цель работы: Приобретение практических навыков в управлении потоками ОС, обеспечение синхронизации между потоками

Задание: Составить и отладить программу на языке Си, осуществляющую работу с процессами и взаимодействие между ними в одной из двух операционных систем. В результате работы программа (основной процесс) должен создать для решение задачи один или несколько дочерних процессов. Взаимодействие между процессами осуществляется через системные сигналы/события и/или каналы (pipe). Необходимо обрабатывать системные ошибки, которые могут возникнуть в результате работы.

Вариант: 8 В файле записаны команды вида: «число число число<endline>». Дочерний процесс производит деление первого числа команда, на последующие числа в команде, а результат выводит в стандартный поток вывода. Если происходит деление на 0, то тогда дочерний и родительский процесс завершают свою работу. Проверка деления на 0 должна осуществляться на стороне дочернего процесса. Числа имеют тип int. Количество чисел может быть произвольным.

Метод решения

Архитектура состоит из 2 процессов parent.exe и child.exe. Они обмениваются данными через канал pipe. Родительский процесс открывает входной файл, создает канал и запускает дочерний процесс, перенаправляя его стандартный ввод на файл, а стандартный вывод — в канал для записи. Сразу после создания потомка родитель закрывает ненужные дескрипторы со своей стороны, чтобы вовремя зафиксировать конец передачи данных. Далее родитель читает результаты из канала до полного завершения вывода, дожидается окончания работы дочернего процесса, получает код его завершения и освобождает все системные ресурсы.

Описание программы

Родительский процесс создаёт анонимный канал, перенаправляет стандартные потоки ввода-вывода дочерней программы, передаёт входные данные из файла, считывает результат работы из канала порциями в буфер и выполняет корректную синхронизацию с освобождением ресурсов ядра.

Структура файлов

Таблица 1 – Файлы проекта

Файл	Назначение
parent.c	Родительский процесс – открывает файл, создает канал, настраивает атрибуты безопасности, запускает дочерний процесс и управляет системными дескрипторами
child.c	Дочерний процесс – читает переданные данные из перенаправленного стандартного ввода и осуществляет их вывод
input.txt	Текстовый файл с входными данными

Основные типы данных и функции

Типы данных и структуры ядра:

HANDLE- дескриптор объекта ядра (файл, канал, процесс, поток).

SECURITY_ATTRIBUTES - параметры безопасности, включает наследование дескрипторов дочерним процессом .

STARTUPINFOA - конфигурация запуска, перенаправляет стандартные потоки ввода-вывода потомка.

PROCESS_INFORMATION- возвращает дескрипторы и идентификаторы созданного процесса и его потока.

DWORD - 32-битное беззнаковое целое число для счётчиков байтов и кодов завершения.

Функции программы:

void PrintError(const char* msg) — принимает строку с описанием ошибки, опрашивает код ошибки через GetLastError() и выводит его в консоль.

int main(void) — точка входа; инициализирует структуры, создаёт канал и файл, порождает дочерний процесс, считывает его вывод, ожидает завершения и освобождает ресурсы ОС.

Используемые системные вызовы

Таблица 2 – Используемые системные вызовы

Системный вызов	Назначение в программе
CreateFileA	Открытие файла <code>input.txt</code> на чтение с правами наследования дескриптора
CreatePipe	Создание анонимного канала в оперативной памяти ядра ОС.
CreateProcessA	Атомарное создание дочернего процесса <code>child.c.exe</code> с передачей перенаправленных дескрипторов ввода-вывода.
ReadFile	Чтение выходных данных из читающего конца канала до появления сигнала EOF
WaitForSingleObject	Блокирующее ожидание полного завершения выполнения дочернего процесса.
GetExitCodeProcess	Считывание кода возврата дочернего процесса для проверки корректности его завершения.
CloseHandle	Заккрытие дескрипторов и освобождение объектов ядра ОС

Ключевые фрагменты кода

Листинг 1 - Создание канала и запрет наследования дескриптора чтения (чтобы канал корректно закрылся после завершения дочернего процесса).

(parent.c)

```
CreatePipe(&pipeRead, &pipeWrite, &saAttr, 0);

si.cb = sizeof(si);
si.dwFlags |= STARTF_USESTDHANDLES;
si.hStdInput = InputFile;
si.hStdOutput = pipeWrite;
si.hStdError = GetStdHandle(STD_ERROR_HANDLE);

CreateProcessA("child.c.exe", NULL, NULL, NULL, TRUE, 0, NULL, NULL, &si, &pi);
```

Листинг 2 - Закрытие дескрипторов записи и потоковое чтение (parent.c)

```
CloseHandle(InputFile);
CloseHandle(pipeWrite);

printf("Results:\n");
while (ReadFile(pipeRead, buffer, sizeof(buffer) - 1, &bytesRead, NULL)) {
    if (bytesRead == 0) break;
    buffer[bytesRead] = '\0';
    printf("%s", buffer);
}
```

Листинг 3 - Синхронизация и освобождение дескрипторов ядра (parent.c)

```
WaitForSingleObject(pi.hProcess, INFINITE);

DWORD exitCode = 0;
if (GetExitCodeProcess(pi.hProcess, &exitCode)) {
    if (exitCode == 2) {
        printf("\nChild process reported\n");
    }
}
CloseHandle(pi.hProcess);
CloseHandle(pi.hThread);
CloseHandle(pipeRead);
```

Результаты

Разработанная программа успешно выполняет чтение входного файла `input.txt`, передаёт данные через анонимный канал в дочерний процесс и выводит обработанный результат в консоль родительского процесса. Также дочерним процессом корректно обрабатывается исключительная ситуация деления на ноль с выводом диагностического сообщения и аварийным завершением по правилам варианта.

Входные данные: input.txt

100 10 60 10 20 2 50 0

Команда запуска и консольный вывод программы:

```
C:\OS_Lab2> parent.c.exe
Results: Error division by zero 0!
10
6
10
Child process reported
Enter for exitДля продолжения нажмите любую клавишу . . .
```

Демонстрация работы strace

Программа запущена под управлением трассировщика системных вызовов ядра ОС Process Monitor; полный вывод трассировки приведён в Приложении Б.

```
[parent.c.exe PID=16748] CreateFile    ...\\input.txt      SUCCESS Desired Access:
Generic Read
```

Родительский процесс открывает файл `input.txt` для передачи его дескриптора на стандартный ввод потомка.

```
[parent.c.exe PID=16748] Process Create ...\\child.c.exe    SUCCESS PID: 22272
```

Системный вызов `CreateProcessA` порождает дочерний процесс `child.c.exe` с новым PID 22272 и перенаправленными потоками ввода-вывода

```
[child.c.exe PID=22272] Process Start          SUCCESS Parent PID: 16748
[child.c.exe PID=22272] Thread Create          SUCCESS Thread ID: 19496
```

Ядро ОС инициализирует адресное пространство потомка и запускает его главный поток выполнения

```
[child.c.exe PID=22272] CloseFile    \\Device\\NamedPipe    SUCCESS
[parent.c.exe PID=16748] CloseFile    \\Device\\NamedPipe    SUCCESS
[child.c.exe PID=22272] Process Exit          SUCCESS Exit Status: 2
```

Потомок вычисляет деление, при делении на 0 выводит ошибку, закрывает канал и завершается вызовом `ExitProcess(2)`. Родитель вычитывает данные, закрывает дескрипторы и завершает работу без утечек ресурсов ядра.

Выводы

В ходе лабораторной работы я изучил, как работают процессы в Windows и как передавать между ними данные через анонимный канал с помощью Win32 API. Я написал две отдельные программы `parent` и `child`, где родитель перенаправляет ввод потомка на файл, а вывод забирает себе через пайп и печатает результаты деления чисел. Работа корректна, потому что данные нормально доходят до конца файла, а при делении на ноль программа не падает, а выводит ошибку и штатно завершается.

Главная сложность была в ошибке с кодом 2 `ERROR_FILE_NOT_FOUND`: сначала родитель не видел файл `input.txt`, а потом не мог найти бинарник `child.c.exe`, потому что Visual Studio запускала проект не из той папки

Приложение А. Исходный код программы

Листинг А.1 – Файл parent.c

```
#include <windows.h>
#include <stdio.h>

#define MAX_PATH_LEN 256
#define BUF_SIZE 512

void PrintError(const char* msg) {
    DWORD errCode = GetLastError();
    printf("%s Error Code: %lu\n", msg, errCode);
}

int main(void) {
    char filename[MAX_PATH_LEN];

    SECURITY_ATTRIBUTES saAttr;

    saAttr.nLength = sizeof(SECURITY_ATTRIBUTES);
    saAttr.lpSecurityDescriptor = NULL;
    saAttr.bInheritHandle = TRUE;

    HANDLE InputFile = CreateFileA("input.txt", GENERIC_READ,
    FILE_SHARE_READ, &saAttr, OPEN_EXISTING, FILE_ATTRIBUTE_NORMAL,
    NULL);

    if (InputFile == INVALID_HANDLE_VALUE) {
        printf("Cant open file Code: %lu\n", GetLastError());
        system("pause");
        return 1;
    }

    HANDLE pipeRead = NULL;
    HANDLE pipeWrite = NULL;

    if (!CreatePipe(&pipeRead, &pipeWrite, &saAttr, 0)) {
        printf("Cant create pipe Code Erore: %lu\n", GetLastError());
        CloseHandle(InputFile);
        system("pause");
        return 1;
    }

    STARTUPINFOA si;
    PROCESS_INFORMATION pi;

    memset(&si, 0, sizeof(si));
    memset(&pi, 0, sizeof(pi));

    si.cb = sizeof(si);
    si.dwFlags |= STARTF_USESTDHANDLES;

    si.hStdInput = InputFile;
    si.hStdOutput = pipeWrite;

    si.hStdError = GetStdHandle(STD_ERROR_HANDLE);

    BOOL success = CreateProcessA("child.c.exe", NULL, NULL, NULL,
    TRUE, 0, NULL, NULL, &si, &pi);

    if (!success) {
        PrintError("Cant start ");
    }
}
```

```

        CloseHandle(InputFile);
        CloseHandle(pipeRead);
        CloseHandle(pipeWrite);
        system("pause");

        return 1;
    }

    CloseHandle(InputFile);
    CloseHandle(pipeWrite);

    char buffer[BUF_SIZE];
    DWORD bytesRead = 0;

    printf("Results:\n");
    while (ReadFile(pipeRead, buffer, sizeof(buffer) - 1, &bytesRead,
        NULL)) {
        if (bytesRead == 0) break;
        buffer[bytesRead] = '\0';
        printf("%s", buffer);
    }

    WaitForSingleObject(pi.hProcess, INFINITE);

    DWORD exitCode = 0;
    if (GetExitCodeProcess(pi.hProcess, &exitCode)) {
        if (exitCode == 2) {
            printf("\nChild process reported\n");
        }
    }

    CloseHandle(pi.hProcess);
    CloseHandle(pi.hThread);
    CloseHandle(pipeRead);

    printf("\n Enter for exit");
    system("pause");

    return 0;
}

```

Листинг А.2 – Файл child.c

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX_LINE 1024

int main(void) {
    char line[MAX_LINE];

    while (fgets(line, sizeof(line), stdin) != NULL) {
        if (strlen(line) == 0) {
            continue;
        }

        int first_num;
        int offset = 0;

        if (sscanf_s(line, "%d%n", &first_num, &offset) != 1) {
            continue;
        }

        int result = first_num;
        int next_num;
        int bytes_read = 0;
        int div_by_zero = 0;

        char* ptr = line + offset;

        while (sscanf_s(ptr, "%d%n", &next_num, &bytes_read) == 1) {
            if (next_num == 0) {
                div_by_zero = 1;
                break;
            }
            result /= next_num;
            ptr += bytes_read;
        }

        if (div_by_zero) {
            fprintf(stderr, "Error division by zero 0! \n");
            fflush(stderr);
            exit(2);
        }

        printf("%d\n", result);
        fflush(stdout);
    }

    return 0;
}
```

Листинг А.3 – Файл Makefile

```
cmake_minimum_required(VERSION 3.10)
project(os_Lab2)

add_executable(parent parent.c.cpp)
add_executable(child child.c.cpp)
```

Приложение Б. Полный вывод strace

Полный вывод *strace* без сокращений для запуска из раздела «Демонстрация работы *strace*»

```
"Time of Day","Process Name","PID","Operation","Path","Result","Detail"
"21:46:51.7673879","parent.c.exe","16748","Process Start","",,"SUCCESS","Parent PID: 4488, Command line: ""C:\Users\Дима\source\repos\parent.c\64\Debug\parent.c.exe"" ,
Current directory: C:\Users\Дима\source\repos\parent.c\64\Debug\, Environment:
;
;
; ACSetupSvcPort=23210
; ALLUSERSPROFILE=C:\ProgramData
; APPDATA=C:\Users\Дима\AppData\Roaming
; CommonProgramFiles=C:\Program Files\Common Files
; CommonProgramFiles(x86)=C:\Program Files (x86)\Common Files
; CommonProgramW6432=C:\Program Files\Common Files
; COMPUTERNAME=HP-OMEN-17
; ComSpec=C:\Windows\system32\cmd.exe
; DriverData=C:\Windows\System32\Drivers\DriverData
; EnableLog=INFO
; FPS_BROWSER_APP_PROFILE_STRING=Internet Explorer
; FPS_BROWSER_USER_PROFILE_STRING=Default
; HOMEDRIVE=C:
; HOMEPATH=\Users\Дима
; JAVA_HOME=C:\Users\Дима\jdk\temurin-21.0.10
; LOCALAPPDATA=C:\Users\Дима\AppData\Local
; LOGONSERVER=\\HP-OMEN-17
; NUMBER_OF_PROCESSORS=12
; OneDrive=C:\Users\Дима\OneDrive
; OS=Windows_NT
; Path=C:\Program Files (x86)\Common
Files\Oracle\Java\javapath;C:\ProgramData\Oracle\Java\javapath;C:\Windows\system32;C:\Windows;C:\Windows\System32\Wbem;C:\Windows\System32\WindowsPowerShell\v1.
0;C:\Windows\System32\OpenSSH;C:\Program Files (x86)\NVIDIA Corporation\PhysX\Common;C:\Program Files\dotnet;C:\Program Files\Microsoft SQL Server\Client
SDK\ODBC\170\Tools\Binn\;C:\Program Files\Microsoft SQL Server\150\Tools\Binn\;C:\Program Files\NVIDIA Corporation\NVIDIA NvDLISR;C:\Program
Files\AmneziaWG\;C:\Program Files\WireGuard\;C:\Program Files\Docker\Docker\resources\bin;C:\Program Files\Git\cmd;C:\Program
Files\Docker\Docker\resources\bin;C:\Program Files\Kubernetes\Minikube\;C:\Program Files\Cloudflare\Cloudflare
WARP\;C:\Users\Дима\AppData\Local\Microsoft\WindowsApps;C:\Users\Дима\dotnet\tools;C:\Users\Дима\AppData\Local\Programs\Microsoft VS
Code\bin;C:\Users\Дима\AppData\Local\Microsoft\WinGet\Links;C:\Users\Дима\AppData\Local\Programs\PyCharm 2025.2.3\bin;C:\Users\Дима\jdk\temurin-
21.0.10\bin;C:\Users\Дима\AppData\Local\Programs\cursor\resources\app\bin;C:\Program Files\Git\cmd;
;
; PATHEXT=.COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH;.MSC
; PROCESSOR_ARCHITECTURE=AMD64
; PROCESSOR_IDENTIFIER=Intel64 Family 6 Model 141 Stepping 1, GenuineIntel
; PROCESSOR_LEVEL=6
; PROCESSOR_REVISION=8d01
;
; ProgramData=C:\ProgramData
; ProgramFiles=C:\Program Files
; ProgramFiles(x86)=C:\Program Files (x86)
; ProgramW6432=C:\Program Files
; PSModulePath=C:\Program Files\WindowsPowerShell\Modules;C:\Windows\system32\WindowsPowerShell\v1.0\Modules
; PUBLIC=C:\Users\Public
; PyCharm=C:\Users\Дима\AppData\Local\Programs\PyCharm 2025.2.3\bin
; RlsSvcPort=22112
; SESSIONNAME=Console
; SystemDrive=C:
; SystemRoot=C:\Windows
; TEMP=C:\Users\1F43~1\AppData\Local\Temp
; TMP=C:\Users\1F43~1\AppData\Local\Temp
; USERDOMAIN=HP-OMEN-17
; USERDOMAIN_ROAMINGPROFILE=HP-OMEN-17
; USERNAME=Дима
; USERPROFILE=C:\Users\Дима
; windir=C:\Windows
; ZES_ENABLE_SYSMAN=1"
"21:46:51.7673970","parent.c.exe","16748","Thread Create","",,"SUCCESS","Thread ID: 27348"
"21:46:51.7714753","parent.c.exe","16748","Load Image","C:\Users\Дима\source\repos\parent.c\64\Debug\parent.c.exe","SUCCESS","Image Base: 0x7ff6b1b70000, Image Size:
0x28000"
"21:46:51.7715678","parent.c.exe","16748","Load Image","C:\Windows\System32\ntdll.dll","SUCCESS","Image Base: 0x7ff9a1eb0000, Image Size: 0x1f8000"
"21:46:51.7721898","parent.c.exe","16748","CreateFile","C:\Users\Дима\source\repos\parent.c\64\Debug","SUCCESS","Desired Access: Execute/Traverse, Synchronize,
Disposition: Open, Options: Directory, Synchronous IO Non-Alert, Attributes: n/a, ShareMode: Read, Write, AllocationSize: n/a, OpenResult: Opened"
"21:46:51.7722813","parent.c.exe","16748","Load Image","C:\Windows\System32\kernel32.dll","SUCCESS","Image Base: 0x7ff9a0300000, Image Size: 0xbff000"
"21:46:51.7723861","parent.c.exe","16748","Load Image","C:\Windows\System32\KernelBase.dll","SUCCESS","Image Base: 0x7ff99f8d0000, Image Size: 0x2f6000"
"21:46:51.7731861","parent.c.exe","16748","CreateFile","C:\Windows\System32\conhost.exe","SUCCESS","Desired Access: Execute/Traverse, Synchronize, Disposition: Open,
Options: Synchronous IO Non-Alert, Non-Directory File, Attributes: N, ShareMode: Read, Delete, AllocationSize: n/a, OpenResult: Opened"
"21:46:51.7732582","parent.c.exe","16748","CreateFileMapping","C:\Windows\System32\conhost.exe","FILE LOCKED WITH ONLY READERS","SyncType:
SyncTypeCreateSection, PageProtection: PAGE_EXECUTE"
"21:46:51.7732843","parent.c.exe","16748","CreateFileMapping","C:\Windows\System32\conhost.exe","SUCCESS","SyncType: SyncTypeOther"
"21:46:51.7733413","parent.c.exe","16748","QuerySecurityFile","C:\Windows\System32\conhost.exe","SUCCESS","Information: Label"
"21:46:51.7733850","parent.c.exe","16748","QueryNameInformationFile","C:\Windows\System32\conhost.exe","SUCCESS","Name: \Windows\System32\conhost.exe"
"21:46:51.7736954","parent.c.exe","16748","QueryEaFile","C:\Windows\System32\conhost.exe","SUCCESS",""
"21:46:51.7737295","parent.c.exe","16748","FileSystemControl","C:\Windows\System32\conhost.exe","OBJECT NOT EXTERNALLY BACKED","Control:
FSCTL_GET_EXTERNAL_BACKING"
"21:46:51.7772289","parent.c.exe","16748","Process Create","C:\Windows\System32\Conhost.exe","SUCCESS","PID: 28404, Command line:
??C:\Windows\system32\conhost.exe 0xfffffff -ForceV1"
"21:46:51.7772641","parent.c.exe","16748","CloseFile","C:\Windows\System32\conhost.exe","SUCCESS",""
"21:46:51.8535870","parent.c.exe","16748","QueryNameInformationFile","C:\Windows\System32\KernelBase.dll","SUCCESS","Name: \Windows\System32\KernelBase.dll"
"21:46:51.8538302","parent.c.exe","16748","QueryNameInformationFile","C:\Windows\System32\KernelBase.dll","SUCCESS","Name: \Windows\System32\KernelBase.dll"
"21:46:51.8539064","parent.c.exe","16748","QueryNameInformationFile","C:\Windows\System32\kernel32.dll","SUCCESS","Name: \Windows\System32\kernel32.dll"
"21:46:51.8543629","parent.c.exe","16748","Thread Create","",,"SUCCESS","Thread ID: 11824"
"21:46:51.8544266","parent.c.exe","16748","Thread Create","",,"SUCCESS","Thread ID: 14632"
"21:46:51.8551779","parent.c.exe","16748","CreateFile","C:\Users\Дима\source\repos\parent.c\64\Debug\ucrtbased.dll","NAME NOT FOUND","Desired Access: Read Attributes,
Disposition: Open, Options: Open Reparse Point, Attributes: n/a, ShareMode: Read, Write, Delete, AllocationSize: n/a"
"21:46:51.8551987","parent.c.exe","16748","CreateFile","C:\Users\Дима\source\repos\parent.c\64\Debug\VCRUNTIME140D.dll","NAME NOT FOUND","Desired Access: Read
Attributes, Disposition: Open, Options: Open Reparse Point, Attributes: n/a, ShareMode: Read, Write, Delete, AllocationSize: n/a"
"21:46:51.8554365","parent.c.exe","16748","CreateFile","C:\Windows\System32\vcruntime140d.dll","SUCCESS","Desired Access: Read Attributes, Disposition: Open, Options:
Open Reparse Point, Attributes: n/a, ShareMode: Read, Write, Delete, AllocationSize: n/a, OpenResult: Opened"
```


"21:46:51,8640059","parent.c.exe","16748","QuerySecurityFile","C:\Users\Дима\source\repos\parent.c\x64\Debug\child.c.exe","SUCCESS","Information: Owner, Group, DACL, SACL, Label, Attribute, Process Trust Label, 0x100"
"21:46:51,8641256","parent.c.exe","16748","CreateFile","C:\Windows\apppatch\sysmain.sdb","SUCCESS","Desired Access: Generic Read, Disposition: Open, Options: Synchronous IO Non-Alert, Non-Directory File, Attributes: N, ShareMode: Read, AllocationSize: n/a, OpenResult: Opened"
"21:46:51,8641812","parent.c.exe","16748","QueryBasicInformationFile","C:\Windows\apppatch\sysmain.sdb","SUCCESS","CreationTime: 28.04.2023 21:21:55, LastAccessTime: 05.10.2026 21:46:51, LastWriteTime: 28.04.2023 21:21:55, ChangeTime: 28.04.2023 21:24:28, FileAttributes: A"
"21:46:51,8641910","parent.c.exe","16748","CloseFile","C:\Windows\apppatch\sysmain.sdb","SUCCESS",""
"21:46:51,8642271","parent.c.exe","16748","QueryBasicInformationFile","C:\Users\Дима\source\repos\parent.c\x64\Debug\child.c.exe","SUCCESS","CreationTime: 15.09.2026 23:45:04, LastAccessTime: 05.10.2026 21:45:38, LastWriteTime: 05.10.2026 19:50:42, ChangeTime: 05.10.2026 19:50:42, FileAttributes: A"
"21:46:51,8653325","parent.c.exe","16748","CloseFile","C:\Users\Дима\source\repos\parent.c\x64\Debug\child.c.exe","SUCCESS",""
"21:46:51,8653757","child.c.exe","22272","Load Image","C:\Users\Дима\source\repos\parent.c\x64\Debug\child.c.exe","SUCCESS","Image Base: 0x7ff6c29d0000, Image Size: 0x27000"
"21:46:51,8654673","child.c.exe","22272","Load Image","C:\Windows\System32\ntdll.dll","SUCCESS","Image Base: 0x7ffa1eb0000, Image Size: 0x1f8000"
"21:46:51,8660382","child.c.exe","22272","CreateFile","C:\Users\Дима\source\repos\parent.c\x64\Debug","SUCCESS","Desired Access: Execute/Traverse, Synchronize, Disposition: Open, Options: Directory, Synchronous IO Non-Alert, Attributes: n/a, ShareMode: Read, Write, AllocationSize: n/a, OpenResult: Opened"
"21:46:51,8661301","child.c.exe","22272","Load Image","C:\Windows\System32\kernel32.dll","SUCCESS","Image Base: 0x7ffa0300000, Image Size: 0xbf000"
"21:46:51,8662527","child.c.exe","22272","Load Image","C:\Windows\System32\KernelBase.dll","SUCCESS","Image Base: 0x7ffa9f8d0000, Image Size: 0x2f6000"
"21:46:51,8672644","child.c.exe","22272","QueryNameInformationFile","C:\Windows\System32\KernelBase.dll","SUCCESS","Name: \Windows\System32\KernelBase.dll"
"21:46:51,8674901","child.c.exe","22272","QueryNameInformationFile","C:\Windows\System32\KernelBase.dll","SUCCESS","Name: \Windows\System32\KernelBase.dll"
"21:46:51,8675794","child.c.exe","22272","QueryNameInformationFile","C:\Windows\System32\kernel32.dll","SUCCESS","Name: \Windows\System32\kernel32.dll"
"21:46:51,8680124","child.c.exe","22272","Thread Create","","SUCCESS","Thread ID: 13504"
"21:46:51,8680929","child.c.exe","22272","Thread Create","","SUCCESS","Thread ID: 26108"
"21:46:51,8685009","child.c.exe","22272","CreateFile","C:\Users\Дима\source\repos\parent.c\x64\Debug\ucrtbased.dll","NAME NOT FOUND","Desired Access: Read Attributes, Disposition: Open, Options: Open Reparse Point, Attributes: n/a, ShareMode: Read, Write, Delete, AllocationSize: n/a"
"21:46:51,8685104","child.c.exe","22272","CreateFile","C:\Users\Дима\source\repos\parent.c\x64\Debug\VCRUNTIME140D.dll","NAME NOT FOUND","Desired Access: Read Attributes, Disposition: Open, Options: Open Reparse Point, Attributes: n/a, ShareMode: Read, Write, Delete, AllocationSize: n/a"
"21:46:51,8690482","child.c.exe","22272","CreateFile","C:\Windows\System32\vcruntime140d.dll","SUCCESS","Desired Access: Read Attributes, Disposition: Open, Options: Open Reparse Point, Attributes: n/a, ShareMode: Read, Write, Delete, AllocationSize: n/a, OpenResult: Opened"
"21:46:51,8690753","child.c.exe","22272","QueryBasicInformationFile","C:\Windows\System32\vcruntime140d.dll","SUCCESS","CreationTime: 05.01.2022 15:34:08, LastAccessTime: 05.10.2026 21:46:51, LastWriteTime: 05.01.2022 15:34:08, ChangeTime: 07.10.2023 21:14:14, FileAttributes: A"
"21:46:51,8690879","child.c.exe","22272","CloseFile","C:\Windows\System32\vcruntime140d.dll","SUCCESS",""
"21:46:51,8691537","child.c.exe","22272","CreateFile","C:\Windows\System32\ucrtbased.dll","SUCCESS","Desired Access: Read Attributes, Disposition: Open, Options: Open Reparse Point, Attributes: n/a, ShareMode: Read, Write, Delete, AllocationSize: n/a, OpenResult: Opened"
"21:46:51,8691867","child.c.exe","22272","QueryBasicInformationFile","C:\Windows\System32\ucrtbased.dll","SUCCESS","CreationTime: 02.12.2020 5:31:00, LastAccessTime: 05.10.2026 21:46:51, LastWriteTime: 02.12.2020 5:31:00, ChangeTime: 07.10.2023 21:14:14, FileAttributes: A"
"21:46:51,8691999","child.c.exe","22272","CloseFile","C:\Windows\System32\ucrtbased.dll","SUCCESS",""
"21:46:51,8692186","child.c.exe","22272","CreateFile","C:\Windows\System32\vcruntime140d.dll","SUCCESS","Desired Access: Read Data/List Directory, Execute/Traverse, Synchronize, Disposition: Open, Options: Synchronous IO Non-Alert, Non-Directory File, Attributes: n/a, ShareMode: Read, Delete, AllocationSize: n/a, OpenResult: Opened"
"21:46:51,8692610","child.c.exe","22272","CreateFileMapping","C:\Windows\System32\vcruntime140d.dll","FILE LOCKED WITH ONLY READERS","SyncType: SyncTypeCreateSection, PageProtection: PAGE_EXECUTE"
"21:46:51,8692876","child.c.exe","22272","CreateFileMapping","C:\Windows\System32\vcruntime140d.dll","SUCCESS","SyncType: SyncTypeOther"
"21:46:51,8692998","child.c.exe","22272","CreateFile","C:\Windows\System32\ucrtbased.dll","SUCCESS","Desired Access: Read Data/List Directory, Execute/Traverse, Synchronize, Disposition: Open, Options: Synchronous IO Non-Alert, Non-Directory File, Attributes: n/a, ShareMode: Read, Delete, AllocationSize: n/a, OpenResult: Opened"
"21:46:51,8693787","child.c.exe","22272","Load Image","C:\Windows\System32\vcruntime140d.dll","SUCCESS","Image Base: 0x7ffa3b200000, Image Size: 0x2b000"
"21:46:51,8694440","child.c.exe","22272","CloseFile","C:\Windows\System32\vcruntime140d.dll","SUCCESS",""
"21:46:51,8694682","child.c.exe","22272","CreateFileMapping","C:\Windows\System32\ucrtbased.dll","FILE LOCKED WITH ONLY READERS","SyncType: SyncTypeCreateSection, PageProtection: PAGE_EXECUTE"
"21:46:51,8694965","child.c.exe","22272","CreateFileMapping","C:\Windows\System32\ucrtbased.dll","SUCCESS","SyncType: SyncTypeOther"
"21:46:51,8695600","child.c.exe","22272","Load Image","C:\Windows\System32\ucrtbased.dll","SUCCESS","Image Base: 0x7ffa380a0000, Image Size: 0x1c7000"
"21:46:51,8696233","child.c.exe","22272","CreateFile","C:\Users\Дима\source\repos\parent.c\x64\Debug\ucrtbased.dll","NAME NOT FOUND","Desired Access: Read Attributes, Disposition: Open, Options: Open Reparse Point, Attributes: n/a, ShareMode: Read, Write, Delete, AllocationSize: n/a"
"21:46:51,8696254","child.c.exe","22272","CloseFile","C:\Windows\System32\ucrtbased.dll","SUCCESS",""
"21:46:51,8697767","child.c.exe","22272","Thread Create","","SUCCESS","Thread ID: 2884"
"21:46:51,8699346","child.c.exe","22272","CreateFile","C:\Windows\System32\ucrtbased.dll","SUCCESS","Desired Access: Read Attributes, Disposition: Open, Options: Open Reparse Point, Attributes: n/a, ShareMode: Read, Write, Delete, AllocationSize: n/a, OpenResult: Opened"
"21:46:51,8699570","child.c.exe","22272","QueryBasicInformationFile","C:\Windows\System32\ucrtbased.dll","SUCCESS","CreationTime: 02.12.2020 5:31:00, LastAccessTime: 05.10.2026 21:46:51, LastWriteTime: 02.12.2020 5:31:00, ChangeTime: 07.10.2023 21:14:14, FileAttributes: A"
"21:46:51,8699747","child.c.exe","22272","CloseFile","C:\Windows\System32\ucrtbased.dll","SUCCESS",""
"21:46:51,8705824","child.c.exe","22272","QueryNameInformationFile","C:\Users\Дима\source\repos\parent.c\x64\Debug\child.c.exe","SUCCESS","Name: \Users\Дима\source\repos\parent.c\x64\Debug\child.c.exe"
"21:46:51,8706238","child.c.exe","22272","ReadFile","C:\Users\Дима\source\repos\parent.c\x64\Debug\input.txt","SUCCESS","Offset: 0, Length: 36, Priority: Normal"
"21:46:51,8729418","child.c.exe","22272","CreateFile","C:\Windows\System32\kernel.appcore.dll","SUCCESS","Desired Access: Read Attributes, Disposition: Open, Options: Open Reparse Point, Attributes: n/a, ShareMode: Read, Write, Delete, AllocationSize: n/a, OpenResult: Opened"
"21:46:51,8729704","child.c.exe","22272","QueryBasicInformationFile","C:\Windows\System32\kernel.appcore.dll","SUCCESS","CreationTime: 06.10.2021 16:30:39, LastAccessTime: 05.10.2026 21:46:51, LastWriteTime: 06.10.2021 16:30:39, ChangeTime: 07.10.2023 21:07:57, FileAttributes: A"
"21:46:51,8729828","child.c.exe","22272","CloseFile","C:\Windows\System32\kernel.appcore.dll","SUCCESS",""
"21:46:51,8730887","child.c.exe","22272","CreateFile","C:\Windows\System32\kernel.appcore.dll","SUCCESS","Desired Access: Read Data/List Directory, Execute/Traverse, Synchronize, Disposition: Open, Options: Synchronous IO Non-Alert, Non-Directory File, Attributes: n/a, ShareMode: Read, Delete, AllocationSize: n/a, OpenResult: Opened"
"21:46:51,8731239","child.c.exe","22272","CreateFileMapping","C:\Windows\System32\kernel.appcore.dll","FILE LOCKED WITH ONLY READERS","SyncType: SyncTypeCreateSection, PageProtection: PAGE_EXECUTE"
"21:46:51,8731745","child.c.exe","22272","CreateFileMapping","C:\Windows\System32\kernel.appcore.dll","SUCCESS","SyncType: SyncTypeOther"
"21:46:51,8732596","child.c.exe","22272","Load Image","C:\Windows\System32\kernel.appcore.dll","SUCCESS","Image Base: 0x7ffa99da0000, Image Size: 0x12000"
"21:46:51,8733747","child.c.exe","22272","Load Image","C:\Windows\System32\msvcr.dll","SUCCESS","Image Base: 0x7ffa00f0000, Image Size: 0x9e000"
"21:46:51,8734845","child.c.exe","22272","Load Image","C:\Windows\System32\prctl.dll","SUCCESS","Image Base: 0x7ffa1d40000, Image Size: 0x126000"
"21:46:51,8736515","child.c.exe","22272","CloseFile","C:\Windows\System32\kernel.appcore.dll","SUCCESS",""
"21:46:51,8741251","child.c.exe","22272","Thread Exit","","SUCCESS","Thread ID: 26108, User Time: 0.0000000, Kernel Time: 0.0000000"
"21:46:51,8741509","child.c.exe","22272","Thread Exit","","SUCCESS","Thread ID: 13504, User Time: 0.0000000, Kernel Time: 0.0000000"
"21:46:51,8741977","child.c.exe","22272","Thread Exit","","SUCCESS","Thread ID: 2884, User Time: 0.0000000, Kernel Time: 0.0000000"
"21:46:51,8753203","child.c.exe","22272","Thread Exit","","SUCCESS","Thread ID: 19496, User Time: 0.0156250, Kernel Time: 0.0000000"
"21:46:51,8755786","child.c.exe","22272","Process Exit","","SUCCESS","Exit Status: 2, User Time: 0.0156250 seconds, Kernel Time: 0.0000000 seconds, Private Bytes: 778 240, Peak Private Bytes: 778 240, Working Set: 3 969 024, Peak Working Set: 3 973 120"
"21:46:51,8756607","child.c.exe","22272","CloseFile","C:\Users\Дима\source\repos\parent.c\x64\Debug","SUCCESS",""
"21:46:51,8758816","child.c.exe","22272","CloseFile","C:\Users\Дима\source\repos\parent.c\x64\Debug\input.txt","SUCCESS",""
"21:46:51,8768413","parent.c.exe","16748","CreateFile","C:\Windows\System32\cmd.exe","SUCCESS","Desired Access: Read Attributes, Disposition: Open, Options: Open Reparse Point, Attributes: n/a, ShareMode: Read, Write, Delete, AllocationSize: n/a, OpenResult: Opened"
"21:46:51,8768793","parent.c.exe","16748","QueryNetworkOpenInformationFile","C:\Windows\System32\cmd.exe","SUCCESS","CreationTime: 06.10.2021 16:30:40, LastAccessTime: 05.10.2026 21:45:48, LastWriteTime: 06.10.2021 16:30:40, ChangeTime: 07.10.2023 21:11:09, AllocationSize: 01.01.1601 3:00:00, EndOfFile: 01.01.1601 3:00:00, FileAttributes: A"
"21:46:51,8768913","parent.c.exe","16748","CloseFile","C:\Windows\System32\cmd.exe","SUCCESS",""
"21:46:51,8770840","parent.c.exe","16748","CreateFile","C:\Windows\System32\cmd.exe","SUCCESS","Desired Access: Read Data/List Directory, Execute/Traverse, Read Attributes, Synchronize, Disposition: Open, Options: Synchronous IO Non-Alert, Non-Directory File, Attributes: N, ShareMode: Read, Delete, AllocationSize: n/a, OpenResult: Opened"
"21:46:51,8771348","parent.c.exe","16748","CreateFileMapping","C:\Windows\System32\cmd.exe","FILE LOCKED WITH ONLY READERS","SyncType: SyncTypeCreateSection, PageProtection: PAGE_EXECUTE"
"21:46:51,8771609","parent.c.exe","16748","CreateFileMapping","C:\Windows\System32\cmd.exe","SUCCESS","SyncType: SyncTypeOther"
"21:46:51,8772283","parent.c.exe","16748","QuerySecurityFile","C:\Windows\System32\cmd.exe","SUCCESS","Information: Label"
"21:46:51,8772781","parent.c.exe","16748","QueryNameInformationFile","C:\Windows\System32\cmd.exe","SUCCESS","Name: \Windows\System32\cmd.exe"
"21:46:51,8775847","parent.c.exe","16748","QueryEaFile","C:\Windows\System32\cmd.exe","SUCCESS",""
"21:46:51,8857878","parent.c.exe","16748","Process Create","C:\Windows\system32\cmd.exe","SUCCESS","PID: 2660, Command line: C:\Windows\system32\cmd.exe /c pause"
"21:46:51,8859783","parent.c.exe","16748","QuerySecurityFile","C:\Windows\System32\cmd.exe","SUCCESS","Information: Owner, Group, DACL, SACL, Label, Attribute, Process Trust Label, 0x100"

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"21:46:51,8860815";"parent.c.exe";"16748";"CreateFile";"C:\Windows\apppatch\sysmain.sdb";"SUCCESS";"Desired Access: Generic Read, Disposition: Open, Options: Synchronous IO Non-Alert, Non-Directory File, Attributes: N, ShareMode: Read, AllocationSize: n/a, OpenResult: Opened"
"21:46:51,8861509";"parent.c.exe";"16748";"QueryBasicInformationFile";"C:\Windows\apppatch\sysmain.sdb";"SUCCESS";"CreationTime: 28.04.2023 21:21:55, LastAccessTime: 05.10.2026 21:46:51, LastWriteTime: 28.04.2023 21:21:55, ChangeTime: 28.04.2023 21:24:28, FileAttributes: A"
"21:46:51,8861610";"parent.c.exe";"16748";"CloseFile";"C:\Windows\apppatch\sysmain.sdb";"SUCCESS";""
"21:46:51,8861971";"parent.c.exe";"16748";"QueryBasicInformationFile";"C:\Windows\System32\cmd.exe";"SUCCESS";"CreationTime: 06.10.2021 16:30:40, LastAccessTime: 05.10.2026 21:45:48, LastWriteTime: 06.10.2021 16:30:40, ChangeTime: 07.10.2023 21:11:09, FileAttributes: A"
"21:46:51,8869003";"parent.c.exe";"16748";"CloseFile";"C:\Windows\System32\cmd.exe";"SUCCESS";""
"21:47:21,8613544";"parent.c.exe";"16748";"Thread Exit";"";"SUCCESS";"Thread ID: 14632, User Time: 0.0000000, Kernel Time: 0.0000000"
"21:47:21,8613633";"parent.c.exe";"16748";"Thread Exit";"";"SUCCESS";"Thread ID: 11824, User Time: 0.0000000, Kernel Time: 0.0000000"
"21:47:21,8614794";"parent.c.exe";"16748";"Thread Exit";"";"SUCCESS";"Thread ID: 22544, User Time: 0.0000000, Kernel Time: 0.0000000"
"21:50:09,6440735";"parent.c.exe";"27904";"Thread Create";"";"SUCCESS";"Thread ID: 25192"
"21:51:51,6068298";"parent.c.exe";"27904";"Thread Exit";"";"SUCCESS";"Thread ID: 25192, User Time: 0.0000000, Kernel Time: 0.0000000"
"21:51:51,8552233";"parent.c.exe";"16748";"Thread Create";"";"SUCCESS";"Thread ID: 29048"
"21:53:33,8152301";"parent.c.exe";"16748";"Thread Exit";"";"SUCCESS";"Thread ID: 29048, User Time: 0.0000000, Kernel Time: 0.0000000"

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